

Research Statement

Project: *A Digital Twin Framework for Behaviourally-Responsive Integrated Passenger-Freight Transport with Crowd-Shipping* **FindAPhD reference:** p196342 **Host:** Aston Business School, Operations & Supply Chain Management Department **Supervisors:** Dr. Seyed Mojtaba Sajadi · Dr. Ammar Al-Bazi **Applicant:** Nauval Zulfikar, MSc Business Analytics (Aston, 2024, 1:1 First Class)

Abstract

Urban transport networks face compounding pressure from urbanisation, e-commerce growth, and the imperative to decarbonise (Cleophas et al., 2019). Integrated passenger-freight transport (IPFT) through crowd-shipping offers a tractable response, using passenger capacity to deliver parcels. Yet long-term viability depends on **behavioural** rather than purely operational factors, trust, willingness, and response to disruption (Punel & Stathopoulos, 2017; Le et al., 2019). Recent literature has advanced Digital Twin (DT) frameworks for city logistics (Belfadel et al., 2023; Tapia & Tavasszy, 2024) and large language model (LLM) applications to transport behaviour (Liu et al., 2024; Nishida et al., 2025; Xu et al., 2025), but no existing work integrates **LLM-generated behavioural decision rules** into a **hybrid agent-based plus discrete-event simulation pipeline** with **explicit longitudinal trust-dynamics modelling under repeated disruption**. This proposal addresses that integration gap. The methodology combines agent-based modelling (ABM) and discrete-event simulation (DES) following the hybrid pattern demonstrated by the proposed supervisors (Sardarzehi, Sajadi & Esmaelnezhad, 2025; Madi, Al-Bazi et al., 2024), with a Mixed-Integer Linear Programming optimisation layer based on the foundational VRPOD formulation (Archetti, Savelsbergh & Speranza, 2016). A cross-context comparative case spanning Birmingham and Bandung is proposed to extend the literature's largely Global-North focus.

1. Introduction and Problem Statement

Cleophas, Cottrill, Ehmke & Tierney (2019) document, in their landscape survey of collaborative urban transportation, that the central obstacles to such systems are not purely technical but *organisational and behavioural*: collaborative initiatives stall on trust, willingness, and the management of shared risk. Crowd-shipping, formalised in the canonical Vehicle Routing Problem with Occasional Drivers (VRPOD) of Archetti, Savelsbergh & Speranza (2016) and further reviewed in Sampaio, Savelsbergh, Veelenturf & Van Woensel (2020), is increasingly proposed as a leverage point for emission and congestion reduction. Tapia, Kourounioti, Thoen & de Bok (2023) develop disaggregate passenger-freight matching models that operationalise this vision.

In parallel, two methodological streams have rapidly matured. First, Digital Twin frameworks for city logistics, Belfadel et al. (2023) propose a conceptual DT framework with 75 citations to date; Tapia & Tavasszy (2024) extend this to freight planning under uncertainty. Second, large language models

have been applied to transport behaviour modelling, Liu et al. (2024) and Nishida, Ishigaki & Onishi (2025) test whether LLMs can predict traveller mode choice, while Alsaleh & Farooq (2025) develop locally-deployable fine-tuned LLMs for mode-choice behaviour and Xu et al. (2025) demonstrate an agentic LLM-driven DT for intermodal freight optimisation.

Despite these advances, four problems remain. **First**, existing DT frameworks are largely static representations of operational state, none explicitly model behavioural variables (trust, willingness) as first-class state variables that evolve over time. **Second**, current LLM applications in transport are one-shot prediction tasks; none integrate LLM-generated rules into a longitudinal simulation pipeline. **Third**, trust dynamics under *repeated* disruption events is under-researched in crowd-shipping (only Le et al., 2019 partially addresses this). **Fourth**, the literature is overwhelmingly Western, no comparative analysis spans Global North and Global South crowd-shipping markets.

2. Research Questions

Three connected research questions structure the work:

RQ1, Architecture. What is the minimum viable architecture for a Digital Twin that integrates the *technical* operations of an IPFT system (vehicle dispatch, parcel matching, terminal queueing) with the *social-behavioural* state of its users (trust, willingness, disruption response)? This question extends Belfadel et al. (2023) by adding an explicit behavioural state layer absent from existing conceptual frameworks.

RQ2, LLM-driven behavioural rules. Can large language models generate adaptive, context-conditioned crowd-shipper decision rules that match or outperform fixed-utility discrete choice baselines (Multinomial Logit, Mixed Logit) under disruption scenarios? This question extends the one-shot LLM mode-choice work (Liu et al., 2024; Nishida et al., 2025; Alsaleh & Farooq, 2025) by embedding LLM-generated rules inside a longitudinal ABM simulation.

RQ3, Disruption resilience and longitudinal trust. How do crowd-shipper trust scores and acceptance elasticities evolve across *repeated* disruption events, and which intervention designs most effectively preserve participation? Le et al. (2019) flag this gap in their supply-side review; no subsequent work has systematically addressed it.

3. Literature Review

Crowd-shipping and integrated passenger-freight transport. Archetti, Savelsbergh & Speranza (2016) introduced the canonical Vehicle Routing Problem with Occasional Drivers (VRPOD), demonstrating that integrating non-employee drivers yields significant cost reductions. Tapia, Kourounioti, Thoen & de Bok (2023) extended this with disaggregate behavioural matching mechanisms. Sampaio et al. (2020) surveyed the broader crowdshipping literature and identified gaps in behavioural heterogeneity modelling. Mohri (2024), in a recent University of Melbourne PhD thesis, examined the

behavioural aspects of crowd-shipping by public transport passengers specifically. None of these works integrates an explicit Digital Twin layer or longitudinal disruption-response modelling.

Digital Twin and LLM applications in transport. The Digital Twin concept, formalised by Grieves & Vickers (2017) in manufacturing, has been extended to city logistics by Belfadel et al. (2023) with a conceptual framework now widely cited. Tapia & Tavasszy (2024) survey Digital Twins for freight planning. In parallel, LLM-based transport behaviour modelling has become an active stream: Liu et al. (2024) assess whether LLMs can capture stated-preference mode choice; Nishida et al. (2025) test LLM mode-choice prediction across alternative sets; Alsaleh & Farooq (2025) develop fine-tuned causal LLMs for mode choice; Xu et al. (2025) demonstrate an agentic LLM-driven DT for intermodal freight. These works are predominantly *single-method*, LLM-as-predictor or LLM-as-orchestrator, without an integrated ABM-DES-MILP simulation pipeline.

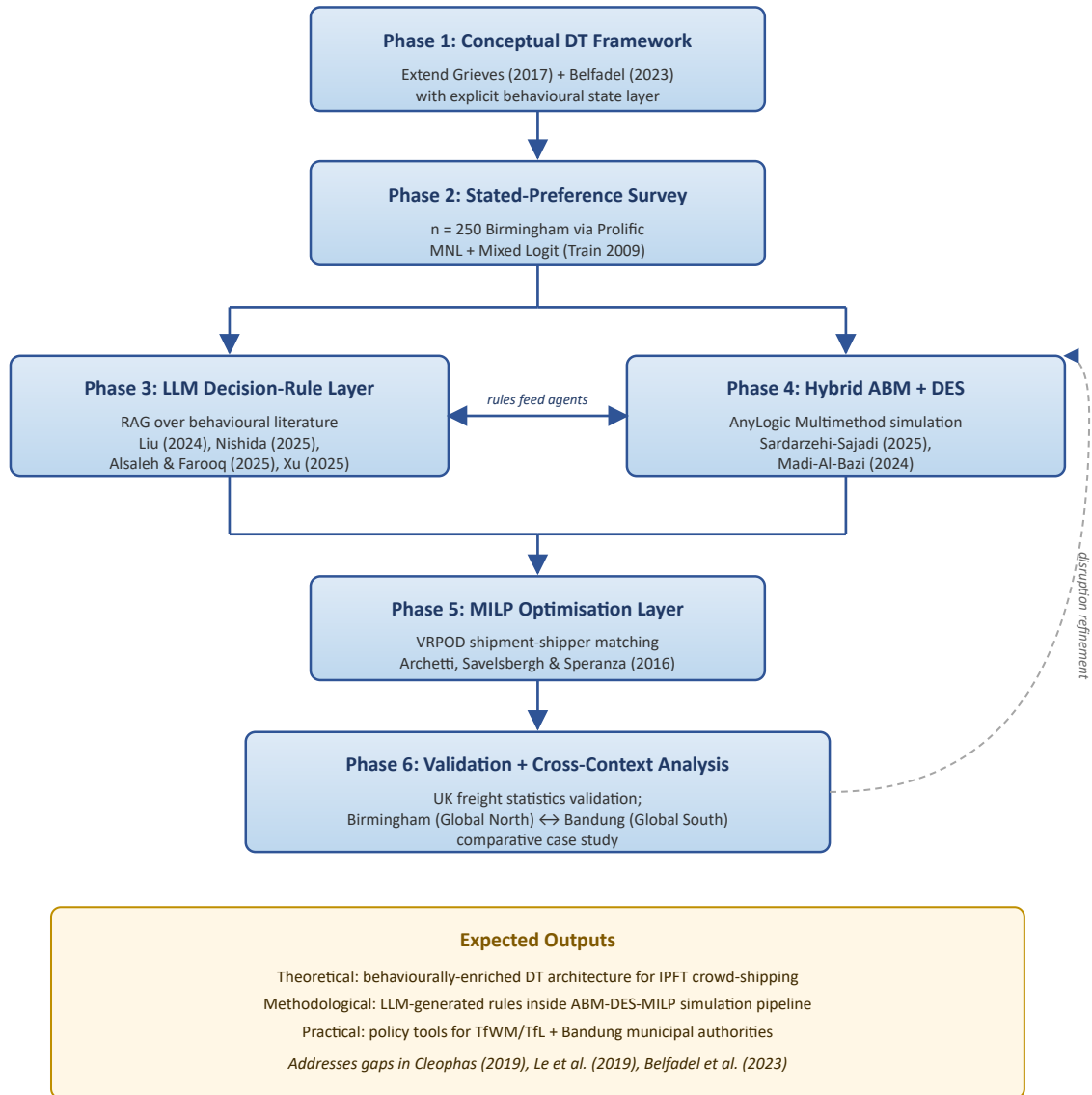
Behavioural and discrete-choice foundations. Ben-Akiva & Lerman (1985) and Train (2009) established the discrete-choice modelling tradition that underpins quantitative behavioural transport research. Punel & Stathopoulos (2017) applied this tradition to crowdsourced delivery acceptance and documented reward saturation in the £25–30 range using stated-preference data from Chicago and Singapore commuters. Le et al. (2019) reviewed crowd-shipping supply-side modelling and identified longitudinal trust dynamics under repeated disruption as an under-researched area. The supervisory pair at Aston has produced exactly the methodological combination this proposal requires: Sardarzehi, Sajadi & Esmaelnezhad (2025) demonstrate Lean Six Sigma integrated with hybrid DES + ABM simulation in the *Journal of Simulation*; Madi, Al-Bazi et al. (2024) develop agent-based heuristics coupled with optimisation in *Soft Computing*.

4. Methodology

4.1 Overview

The research proceeds through six interconnected phases. A conceptual DT framework (Phase 1) extends Grieves & Vickers (2017) and Belfadel et al. (2023) by adding an explicit behavioural state layer. Empirical behavioural data is collected through a stated-preference (SP) survey (Phase 2). LLM-generated decision rules (Phase 3) and agent-based plus discrete-event simulation (Phase 4) compose the behaviourally-responsive simulation engine. An MILP optimisation layer (Phase 5) follows the Archetti et al. (2016) VRPOD formulation. Cross-context validation and policy synthesis (Phase 6) closes the cycle. The full workflow and inter-phase data flow are summarised in Figure 1.

Figure 1. Six-phase methodology workflow



Bidirectional arrow (Phase 3 ↔ Phase 4): LLM-generated decision rules parametrise ABM agents. Dashed feedback: disruption-induced rule refinement.

4.2 Quantitative Methods

Quantitative methods include stated-preference survey design and deployment via Prolific (n = 200–500 Birmingham commuters), Multinomial Logit and Mixed Logit estimation in Biogeme following Train (2009), and Mixed-Integer Linear Programming optimisation for shipment–shipper matching following Archetti, Savelsbergh & Speranza (2016). The agent-based simulation will be implemented in AnyLogic Multimethod combined with discrete-event modules, following the hybrid methodology demonstrated by Sardarzehi, Sajadi & Esmaelnezhad (2025) and Madi, Al-Bazi et al. (2024). A 450-run parameter sweep prototype (already implemented, see Section 6) demonstrates the saturation pattern Punel & Stathopoulos (2017) report.

4.3 Qualitative Methods

Qualitative methods focus on LLM-driven analysis of crowd-shipper decision-making. Building on Liu et al. (2024), Nishida et al. (2025), and Alsaleh & Farooq (2025), but moving beyond one-shot prediction, this work will use retrieval-augmented generation (RAG) over a curated behavioural-literature knowledge base to produce *adaptive*, context-conditioned decision rules per traveller profile. This methodology continues the LLM-blockchain-supply-chain work of my MSc dissertation, supervised by Dr. Al-Bazi (2024).

4.4 Interplay Between Quantitative and Qualitative Methods

The integration of quantitative and qualitative methods is the project's central methodological contribution. LLM-generated decision rules (qualitative) parametrise agent heterogeneity in the ABM (quantitative). SP-survey-estimated Mixed Logit baselines (quantitative) provide a falsifiable benchmark for LLM-generated rules (qualitative). Disruption-induced trust changes (qualitative narrative) are quantitatively measured through acceptance-rate decay across repeated scenarios. This synergy distinguishes the project from LLM-as-predictor work (Liu et al., 2024; Nishida et al., 2025) and from agentic-LLM-as-orchestrator work (Xu et al., 2025).

5. Expected Outcomes and Significance

The project will deliver three primary contributions. **Theoretically**, a behaviourally-enriched Digital Twin architecture for IPFT crowd-shipping that explicitly models trust dynamics under repeated disruption, addressing the gap identified by Le et al. (2019). **Methodologically**, a demonstrated integration of LLM-generated adaptive decision rules into a hybrid ABM-DES-MILP simulation pipeline, extending the one-shot LLM mode-choice work of Liu et al. (2024), Nishida et al. (2025), and Alsaleh & Farooq (2025), and complementing the agentic-LLM-DT freight optimisation of Xu et al. (2025) with a passenger-freight crowd-shipping focus. **Practically**, deployable policy tools for crowd-shipping operators (Transport for West Midlands, Transport for London, Indonesian municipal authorities) that quantify the cost-effective reward threshold, disruption resilience envelope, and intervention efficacy. The cross-context Birmingham–Bandung comparative analysis extends the literature beyond its current Western-centric focus (Cleophas et al., 2019; Punel & Stathopoulos, 2017).

6. Fit with Existing Background

Four pre-application prototypes demonstrate methodological readiness, each available publicly:

- **Mini Digital Twin** (github.com/nauvalZulfikar/Mini-Digital-Twin): three-layer FastAPI
 - mesa + simpy + Streamlit DT prototype with interactive disruption injection. This is the architectural skeleton of Phase 1.

- **LLM Shipper Profiles** (github.com/nauvalZulfikar/LLM-Shipper-Profiles-): end-to-end RAG + gpt-4o-mini decision-rule generator producing literature-grounded DecisionRule objects. Direct continuation of my November 2024 CV-Job Suitability LLM work and my MSc dissertation under Dr. Al-Bazi.
- **VRPOD MILP** (github.com/nauvalZulfikar/VRP-MILP): Archetti et al. (2016) replication solved with scipy.optimize.milp. Crowd-shipping integration reduces cost by **38.5%** on a toy 8-customer instance.
- **Crowd-Shipping ABM Sweep** (github.com/nauvalZulfikar/Crowd-Shipping-ABM-): 450-run parameter sweep with auto-generated findings citing Punel & Stathopoulos (2017) and Le et al. (2019). Supply effect: 45.34% (n_{pax} = 20) → 79.80% (n_{pax} = 100).

The MSc dissertation, *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach*, supervised by Dr. Al-Bazi at Aston, establishes my direct methodological lineage to the LLM-driven analysis component of this PhD.

7. Three-Year Workplan

Year 1: Foundations. Structured literature review (extending the 18-reference base already assembled). SP survey design and deployment via Prolific (n ≈ 250 Birmingham respondents). MNL and Mixed Logit baseline estimation in Biogeme (Train, 2009). First formal ABM in AnyLogic on a small Birmingham network, extending the mesa prototype already public. Target: workshop submission to the UK Transport Study Group (UTSG) or the Operational Research Society (ORS) annual conference.

Year 2: Integration and disruption modelling. Full Birmingham IPFT scenario with GTFS data and crowd-shipper supply model from Le et al. (2019). LLM decision-rule layer integrated into the ABM, benchmarked against the Mixed Logit baseline from Year 1. Disruption scenario library covering weather, breakdown, surge, and *repeated-disruption sequences*, the longitudinal trust dimension. Target: journal submission to *Transportation Research Part E* or *Transport Policy*.

Year 3: Optimisation, comparative analysis, and synthesis. MILP optimisation layer integrated with the DT for closed-loop shipment–shipper matching, extending the static Archetti et al. (2016) formulation. Bandung comparative case study, leveraging field network from prior employment at the Bandung Public Works and Spatial Planning Department. Synthesis paper targeting *Transportation Science* or the *Journal of Operations Management*. Thesis writing.

8. Challenges and Limitations

Three principal challenges are anticipated, each with a mitigation strategy. **First**, formal simulation modelling in AnyLogic, an essential requirement of the project, is not yet a demonstrated strength of

my profile. Mitigation: I have begun AnyLogic Personal Learning Edition tutorials and have planned a three-month structured ramp-up programme; the Python mesa

- simply prototype already publicly available demonstrates competence in the underlying ABM and DES paradigms. **Second**, LLM-generated decision rules carry hallucination risk; the literature notes opacity in LLM outputs (Liu et al., 2024). Mitigation: rules will be validated against the Mixed Logit baseline (Year 2), and a deterministic rule-based fallback grounded in the same behavioural literature is already implemented in the public prototype. **Third**, the Bandung comparative case raises questions of comparability, different network topology, fare structures, and cultural norms. Mitigation: the comparison will be framed analytically, not as parameter transfer, with explicit acknowledgement of scope conditions.
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9. Conclusion and Why Aston

Crowd-shipping cannot be evaluated solely on operational efficiency: it succeeds or fails on trust, willingness, and behavioural response to disruption. The proposed Digital Twin framework operationalises this insight by integrating LLM-generated decision rules into a hybrid ABM-DES-MILP simulation pipeline, with longitudinal trust dynamics under repeated disruption as its central novel contribution. Aston Business School is the right home for this work. Dr. Sajadi and Dr. Al-Bazi jointly provide the methodological coverage, hybrid DES + ABM (Sardarzehi, Sajadi & Esmaelnezhad, 2025) plus agent-based heuristics with optimisation (Madi, Al-Bazi et al., 2024), that the project requires. As an Aston alumnus returning with a 1:1 First Class MSc, an Aston Enterprise Scholarship, and a dissertation directly supervised by Dr. Al-Bazi on the LLM-supply-chain methodology now central to this proposal, I bring institutional fluency, prior research engagement, and methodological continuity that few applicants can match.

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